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REVIEW – CLINICAL ONCOLOGY

T-cell-associated cellular immunotherapy for lung cancer

Ke Li · Qing Zhang · Yang Zhang · Jie Yang · Junnian Zheng

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Abstract
Purpose The aim of the present study was to discuss recent findings on the role of T cells in lung cancer to provide information on their potential application, especially in cellular immunotherapy.
Methods Data on the different types of T cells that are currently used for the treatment of lung cancer were obtained by searching the PUBMED database.
Results Cytotoxic T lymphocytes, natural killer T cells, γδ T cells, lymphokine-activated killer cells, tumor-infiltrating lymphocytes, cytokine-induced killer cells and gene-modified T cells were analyzed to determine the benefits and drawbacks of their application in the treatment of lung cancer. Advances in the study of their antitumor mechanisms and directions for future research were discussed.
Conclusions T cells are critical for tumorigenesis and therefore important targets for the treatment of lung cancer. T-cell-associated cellular immunotherapy opens up a window of opportunity for the development of complementary methods to traditional lung cancer treatments, which warrants further investigation to improve the clinical outcomes of lung cancer patients.

Keywords T cell · Lung cancer · Treatment · Cellular immunotherapy

Ke Li and Qing Zhang have contributed equally to this paper.

K. Li · Q. Zhang · Y. Zhang · J. Yang · J. Zheng (✉)
Jiangsu Key Laboratory of Biological Cancer Therapy, Xuzhou Medical College, West Huaihai Road 84#, Xuzhou 221002, Jiangsu, China
e-mail: jnzheng@xzmc.edu.cn

Introduction

Lung cancer is the most common cancer and the leading cause of cancer mortality (followed by prostate cancer and breast cancer) among women and men worldwide, accounting for approximately 1.3 million deaths each year (Polo and Besse 2013; Siegel et al. 2013). It is characterized by high incidence, with approximately 228,000 newly diagnosed cases in the USA in 2013 (Siegel et al. 2013), and poor prognosis. Current treatments for lung cancer include surgery, chemotherapy and radiotherapy. In the last two decades, different tumor immunotherapy techniques have been developed that have shown clinical benefits, especially in the area of cellular immunotherapy. T cells, the core of cellular immunotherapy, play an important role in the study and treatment of lung cancer.

With the help of major histocompatibility complex (MHC) class I molecules, tumor cells present antigens to $CD8^+$ T cells and activate them, which requires not only an antigen-specific signal mediated by the T-cell receptor (TCR) but also co-stimulatory signals. The activated $CD8^+$ T cells release cytokines to kill tumor cells concomitant with the $CD4^+$ Th-cell-mediated secretion of interleukin 2 (IL-2) and interferon (IFN)-γ to enhance the effect of $CD8^+$ T cells on tumor cells (Fig. 1). Immune escape pathways allow some tumor cells to survive. Inhibitory cytokines secreted by tumor cells, inhibitory regulatory T cells and the downregulation of MHC molecules, co-stimulatory molecules and tumor-associated antigens (TAAs) weaken T cells and decrease their ability to eliminate tumor cells. To enhance the antitumor effects of T cells, their quantity or quality should be increased by promoting their amplification or by improving their ability to recognize and attack tumor cells.

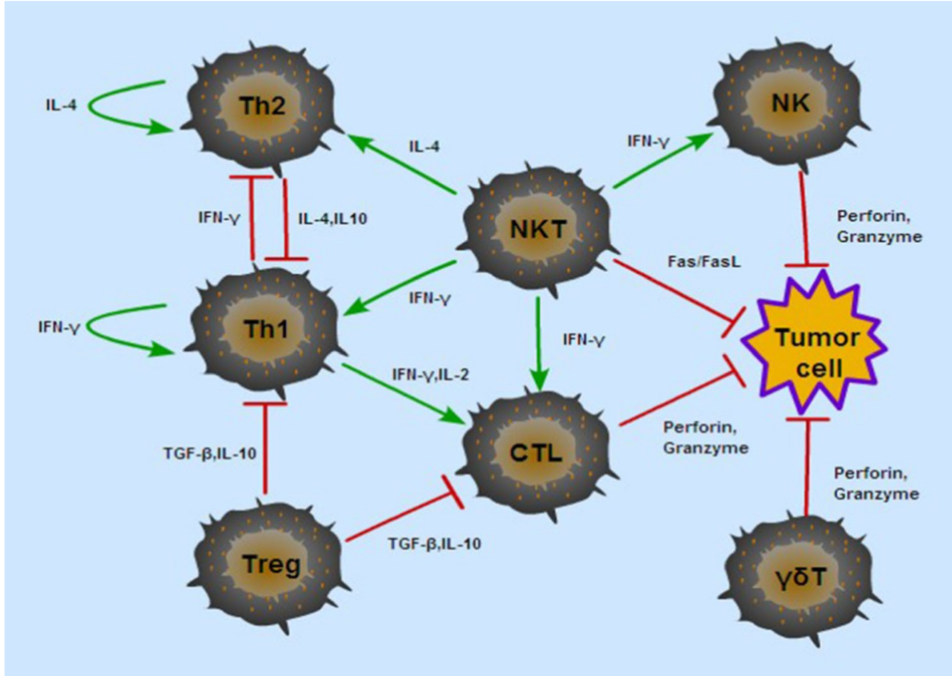

Fig. 1 Interactions between T-cell subtypes and tumor cells

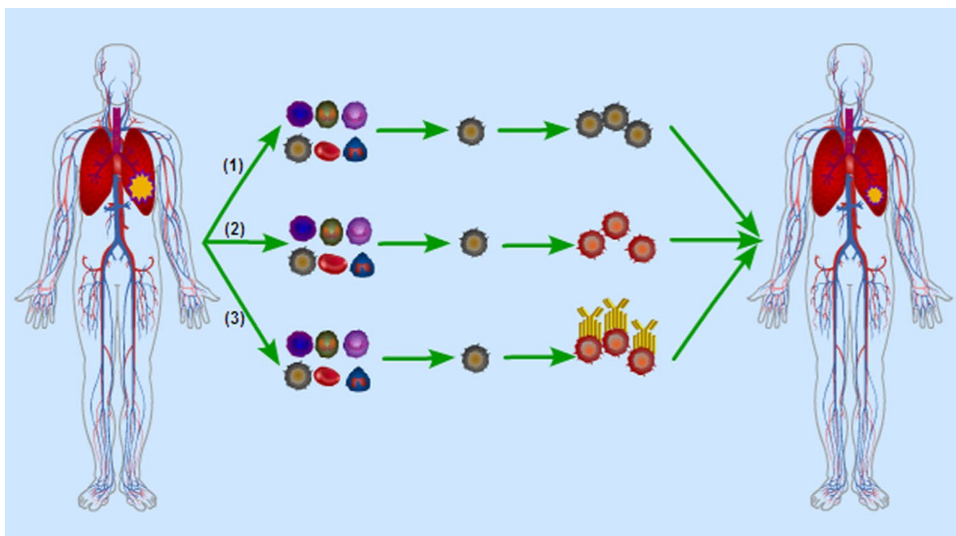

Fig. 2 Common methods of cellular immunotherapy for lung cancer

Several clinical trials have been conducted to test different immunotherapy strategies for the treatment of lung cancer, and most of them have shown encouraging results. Three methods are commonly used in clinical practice as follows: (1) separating lymphocytes from the peripheral blood of patients and re-infusing the lymphocytes after proliferation in vitro (e.g., γδ T cells); (2) separating lymphocytes from peripheral blood, tumor tissues or lymph nodes of patients and re-infusing the activated lymphocytes after proliferation and activation in vitro [e.g., cytokine-induced killer (CIK) cells]; (3) separating lymphocytes from the peripheral blood of patients and re-infusing gene-modified lymphocytes after proliferation and modification in vitro [e.g., chimeric antigen receptor (CAR)-modified T cells] (Fig. 2).

Cytotoxic T lymphocytes

Cytotoxic T lymphocytes (CTLs) are $CD8^+$ αβ T cells, which are the main force of antitumor immunity. These cells have the ability to recognize MHC class I molecules, presenting TAAs and releasing granzymes and perforins to lyse tumor cells (Fig. 1). The number of $CD8^+$ T cells in cancer nests and in the stroma is an independent prognostic factor in stage IV non-small cell lung cancer (NSCLC) patients treated with chemotherapy (Kawai et al. 2008). Hence, methods to increase the number of CTLs are being developed.

The downregulation of the expression of MHC class I molecules is found in 25–94 % of NSCLCs (Baba et al. 2007), leading to the inability of CTLs to recognize tumor cells. Baba et al. argued that restoration of MHC class I chain-related molecule A expression may abrogate evasion from cellular attack by CTLs and natural killer (NK) cells. Another strategy that may turn out to be fruitful is the application of antibodies. Programmed cell death-1 (PD-1) and cytotoxic T lymphocyte antigen-4 (CTLA-4) are expressed on the surface of T cells, and they are related to the immune incompetence of T cells. Their antibodies have been shown to be clinically effective for the treatment of lung cancer (Aerts and Hegmans 2013; Topalian et al. 2012).

Reinfused CTLs are not effective because of the downregulation of MHC and co-stimulatory molecules. Under these circumstances, there are no successful cases of treatment of lung cancer by CTL reinfusion. The identification of crucial factors potentially associated with the loss of MHC expression may provide a new direction for the development of immunotherapy strategies (Baba et al. 2010). Injection of peptide vaccines, which are similar to TAAs, stimulates T cells, resulting in an increase in the number of CTLs available to attack tumor cells whose TAAs are positive. Phase II or III clinical trials testing the effect of lung cancer vaccines are ongoing according to the website www.clinicaltrials.gov. Cancer vaccines were developed to stimulate antitumor immune responses, and their use in combination with traditional treatment methods may improve patient outcomes.

Natural killer T cells

NKT cells, which were first identified in 1986, share properties of both T cells and NK cells. NKT cells are characterized by the expression of unique invariant T-cell receptors encoded by Vα14-Jα18 gene segments in mice and Vα24-Jα18 in humans (Fujii et al. 2013), and they can recognize α-galactosylceramide (α-GalCer) presented by CD1d. Invariant NKT (iNKT) cells increase antitumor responses because they can rapidly produce large amounts of IFN-γ, which is a hallmark of inflammatory cytokines and is critical for NK cells to attack MHC-negative tumors and for CTLs to kill MHC-positive tumors (Fig. 1).

Jusufovic et al. (2011) reported that NKT cell responses are more active in cancerous lung tissues than in the healthy lung of the same patient and in small cell lung cancer (SCLC) than in NSCLC. Additionally, these reactions are induced with an increase in the clinical stage. Decreased numbers and reduced function of iNKT cells are observed in patients with various malignant cancers and in correlation with poor clinical outcomes (Motohashi et al. 2011). Therefore, therapeutic intervention strategies aimed at the recovery of functional NKT cells would be appropriate methods for the treatment of cancer. In a phase I study, activated NKT cell administration was well tolerated and performed safely with minor adverse effects in patients with advanced and recurrent NSCLC ($n = 6$) (Motohashi et al. 2006). A clinical trial of iNKT cell-based immunotherapy showed that the infusion of ligand-pulsed antigen-presenting cells and/or activated iNKT cells is safe and well tolerated (Motohashi et al. 2011). The administration of α-GalCer-pulsed APCs was shown to induce a dramatic infiltration and activation of iNKT cells in the tumor microenvironment and augment IFN-γ production by the α-GalCer-stimulated TILs (Nagato et al. 2012). Similarly, in patients with advanced and recurrent NSCLC, α-GalCer-pulsed IL-2/GM-CSF-cultured peripheral blood mononuclear cells (PBMCs) were associated with prolonged median survival time (MST) in a phase I–II study (Motohashi et al. 2009). Therefore, this immunotherapy technique warrants further evaluation (Table 1).

Currently, there are two research hotspots in the field of NKT cells. One is induced pluripotent stem (iPS)-derived NKT cells. iPS cells are generated from mature NKT cells using Oct3/4, Sox2, Klf4, and c-Myc genes and then develop into functional NKT cells in vitro in the presence of IL-7 and Flt3L (Fujii et al. 2013). The iPS-derived NKT cells are able to function well in vivo, but further research is necessary before their clinical application. The other one is a vector cell containing tumor mRNA and α-GalCer/CD1d (Fujii et al. 2013), which is designed to promote the induction of tumor-specific long-term memory $CD8^+$ T-cell responses and thus tumor growth inhibition. These two types of cells may be new candidates for the treatment of lung cancer.

γδ T cells

γδ T cells account for 2–10 % of T lymphocytes in human blood and play a role in immune surveillance. Unlike αβ T cells, whose activation requires antigen processing and MHC-restricted peptides displayed by antigen-presenting cells, γδ T cells recognize tumor antigens directly through the γδ TCR and exhibit potent MHC-unrestricted lytic activity against microbial pathogens and tumors.

Clinical trials have been conducted to evaluate the safety and efficacy of γδ T-cell-based immunotherapy for non-Hodgkin lymphoma, multiple myeloma and solid tumors

Table 1 Comparison among different kinds of adoptive cellular immunotherapy for lung cancer

Categories	Patient no.	Types	Stage	Culture period	Average infused cell no.	Clinical results	References
NKT	17	NSCLC	rec./IIIb–IV	1–2 weeks	6.2×10^9	SD in 5, PD in 12, MST↑	Motohashi et al. (2009)
γδT	10	NSCLC	rec.	2 weeks	9.73×10^9	SD in 3, PD in 5	Nakajima et al. (2010)
	15	NSCLC	rec./IIIb–IV	2 weeks	1.57×10^{10}	SD in 6, PD in 6	Sakamoto et al. (2011)
LAK	121	NSCLC/SCLC	IIIb–IV	4–6 days	3.7×10^9	ORR↑	Liu et al. (1993)
	51	NSCLC	I–IV	2–3 weeks	7.74×10^9	SR↑	Kimura and Yamaguchi (1995)
	82	NSCLC	I–IV	2–3 weeks	9.7×10^9	SR↑	Kimura and Yamaguchi (1997)
	19	NSCLC	I	12–14 days	7.4×10^9	ORR–	Yano et al. (1999)
TIL	11	NSCLC	IIIb	5–6 weeks	2.34×10^{10}	SR↑	Melioli et al. (1996)
	56	NSCLC	II–III	6–8 weeks	2×10^{10}	SR↑	Ratto et al. (1996)
	10	NSCLC/SCLC	IIIb–IV	2–3 weeks	3.5×10^{10}	ORR↑	Tan et al. (1996)
CIK	29	NSCLC	IIIb–IV	2 weeks	5.0×10^9	ORR↑, DCR↑, TP↑, PFS↑, OS↑	Wu et al. (2008)
	42	NSCLC	I–IIIa	2 weeks	1.04×10^{11}	SR↑	Li et al. (2009)
△	14	NSCLC	IIIb–IV	9 days	$3.24 \times 10^7 + 5.32 \times 10^9$	TTP↑, SR↑	Zhong et al. (2011)
	87	NSCLC	I–IV	2 weeks	5.23×10^{10}	OS↑, PFS↑	Li et al. (2012)
	85	NSCLC	IV	15–16 days	4×10^{10}	OS↑	Yuanying et al. (2013)
△	30	NSCLC	IIIb–IV	2 weeks	$8.5 \times 10^6 + 1.35 \times 10^9$	OS↑, TTP↑	Zhong et al. (2014)
	61	NSCLC	IIIb–IV	2 weeks	1.0×10^{11}	OS↑	Yang et al. (2013)

SD stable disease, *PD* progressive disease, *MST* median survival time, *ORR* overall response rate, *SR* survival rate, *DCR* disease control rate, *TTP* time to progression, *PFS* progression-free survival, *OS* overall survival

△, DC/CIK; ↑, results improved compared with control groups or the ones reported in literature; –, not different from control groups

including lung cancer (Yoshida et al. 2011). The most widely used are the Vγ9δ2 T cells, which are particularly effective at cross-presenting soluble proteins to $CD8^+$ T cells (Brandes et al. 2009). The synthetic agonist phosphostim and IL-2 can expand Vγ9δ2 T cells from multiple myeloma patients and promote the efficient and stable killing of human myeloma cells (Burjanadze et al. 2007). A similar method was shown to be feasible for the treatment of renal cell carcinoma (Viey et al. 2005). Two phase I studies confirmed the safety and effectiveness of γδ T cells in lung cancer patients (Nakajima et al. 2010; Sakamoto et al. 2011) (Table 1).

Current strategies aimed at improving the antitumor effects of γδ T cells are based on combination with bisphosphonates and gene modification (Li 2013). The former can improve tumor immune surveillance by stimulating the amplification of γδ T cells (Russell 2011), and the latter is aimed at enhancing recognition and attack (Hanagiri et al. 2012). Co-transduction of αβ TCR and CD8 into γδ T cells yields the same antigen-specific activity as that of original CTLs in vitro and in vivo (Hanagiri et al. 2012). The transfer of genetically modified γδ T cells resulted in the regression of metastatic melanoma (Morgan et al. 2006). Meanwhile, CDR3δ-grafted-γ9δ2 T cells adoptively transferred into nude mice have antitumor effects on ovarian carcinoma (Zhao et al. 2012). These results indicate that modified γδ T cells may be developed as a therapeutic strategy for the treatment of lung cancer.

Lymphokine-activated killer cells

Lymphokine-activated killer (LAK) cells were first reported in 1982 by Rosenberg's laboratory (Grimm et al. 1982). In vitro, lymphocytes can be stimulated by IL-2 to kill tumor cells insensitive to CTLs or NK cells. The application of LAK cells in the treatment of advanced tumors can be traced to 1985.

To enhance the power of LAK cells, patients receive recombinant IL-2 (rIL-2) during the treatment. Intrapleural transfer of autologous or allogeneic LAK cells combined with rIL-2 was used in the treatment of 121 patients with malignant effusion associated with advanced lung cancer. The effusion disappeared in 71 patients (58.6 %) and was significantly decreased in 45 patients (36.2 %), and no serious side effect was observed (Liu et al. 1993). In another clinical trial, 105 patients who had undergone non-curative resection of primary lung cancer were divided into two groups randomly. The group receiving IL-2 and LAK cells combined with radiation therapy or chemotherapy showed a better 7-year survival rate than the control group, in which patients were treated by radiation therapy or chemotherapy (Kimura and Yamaguchi 1995). A similar result was obtained in a phase III study (Kimura and Yamaguchi 1997). Both granulocyte macrophage colony stimulating factor (GM-CSF) and high-dose glucocorticoids increase LAK activity in lung cancer patients (Takahashi et al. 1995; Kato et al. 2010).

Because of the large amount of IL-2 associated with the clinical application of LAK cells, serious side effects have been reported such as capillary leak syndrome (CLS), which can lead to hypotension, oliguria, pulmonary edema and dyspnea (Atkins et al. 1999, 2000; Rosenberg et al. 1994). These effects constitute an important obstacle limiting the development of LAK cells for clinical application.

Tumor-infiltrating lymphocytes

Tumor-infiltrating lymphocytes (TILs), which were first identified in 1986 by Rosenberg's group, are isolated from tumor samples, draining lymph nodes or malignant effusion. One specimen is processed into a single cell suspension that is exposed to high-dose IL-2 to increase the number of lymphocytes, which are then reinfused into the patient. Tumor stroma inflammatory cells are mainly TILs (approximately 2/3) and tumor-associated macrophages (TAMs, approximately 1/3), but they are poorly replicating and mainly recruited to the tumor stroma (Kataki et al. 2002).

Studies have shown the effectiveness of TIL therapy in NSCLC. High levels of intratumoral TILs are associated with a decreased risk of disease recurrence and improved disease-free survival (Kilic et al. 2011; Horne et al. 2011). The infusion of in vitro expanded TILs derived from surgical samples is feasible and was shown to prolong overall survival and control residual disease in patients with advanced NSCLC (Melioli et al. 1996). In 1996, TILs engineered by the IL-2 gene were reinfused into ten advanced lung cancer patients with pleural effusions. The pleural effusions did not reaccumulate for at least 4 weeks in six patients, and the size of the original tumor decreased in one patient (Tan et al. 1996).

Reinfusion of TILs into lung cancer patients has certain advantages such as its specificity and safety. However, limitations include the difficulty in obtaining samples from surgeries, adverse effects associated with the combination with high-dose IL-2 and a long culture period such as 5 weeks (Table 1). In contrast to the modest success of cell transfer therapy for melanoma, clinical experience in lung cancer has been far from satisfactory (Goff et al. 2010).

Cytokine-induced killer cells

Cytokine-induced killer (CIK) cells were first reported in 1991 by Schmidt Wolf of Stanford University. Stimulation with different kinds of cytokines, such as CD3 monoclonal antibodies, IL-2 and IFN-γ, results in the conversion of PBMCs into CIK cells with the characteristic $CD3^{+}CD56^{+}$ phenotype.

The advent of CIK cells marked a turning point in adoptive therapy methodology. Many clinical trials have been conducted worldwide in the last two decades, especially in China. In a phase II clinical study, autologous CIK cell immunotherapy improved the efficacy of conventional chemotherapy in advanced stage NSCLC patients (Li et al. 2012). DC-activated CIK cells enhance antitumor effects, and chemotherapy combined with CIK/DC cells can improve the clinical outcomes of advanced NSCLC patients (Yang et al. 2013). Furthermore, combination treatment with CIK cells and endostatin or DC-based cancer vaccines may have a synergistic effect on improving clinical outcomes (Shi et al. 2013; Cui et al. 2013).

In recent years, CIK cells have been widely used as immunotherapy for many cancers because of their high proliferation rate and cytotoxic activity, especially after activation by DCs. Approximately 2 weeks are needed to culture cells in vitro until reinfusion except in TILs (Table 1). The number of CIK cells reaches a peak from days 14 to 21 after being induced by cytokines, increasing by about 600-fold on average, coinciding with the highest cytotoxic activity of the CIK cells (Wu et al. 2008). The cumulative number of transferred CIK cells ranges from 9 to 11 orders of magnitude, whereas the simultaneous transfer of DCs can enhance the function of CIK cells and reduce the total number of cells (Table 1). Therefore, CIK (CIK/DC) therapy is widely used because of its simplicity, moderate culture period, few adverse effects and satisfactory compliance of patients (Zhong et al. 2014; Jin et al. 2013; Shi et al. 2012). However, CIK therapy is currently not a routine treatment for lung cancer such as surgery, chemotherapy and radiation. There are two possible explanations for this as follows: (1) Unlike drugs that can be produced on a large scale, cell therapy needs to be individualized; and (2) the current cost of immunotherapy is high, and some patients cannot afford it. Therefore, reducing the cost of CIK therapy could make it available to a larger number of patients.

Gene-modified T cells

The first legal gene transfer studies were approved 20 years ago, and gene therapy for cancer remains promising. Tumor cells express a certain amount of TAAs on the surface, which are important for their identification by T cells. T cells engineered to recognize TAAs expressed on the surface of tumor cells can circumvent local and systemic tolerance mechanisms that limit endogenous antitumor T cells and can overcome the logistical difficulties in isolating and expanding tumor-reactive T cells from patients. The MHC-restricted αβ TCR gene has been widely studied, but its effectiveness for the treatment of lung cancer has only been assessed in animal models. In addition, the advent of hybrid TCRs has slowed its rate of development. For example, chimeric antigen receptor (CAR)-modified T cells are the latest form of gene therapy. These receptors couple a major MHC-unrestricted interaction to the signaling mechanism of activating T cells.

The CAR intracellular signaling domain is modified from a single signaling unit to costimulatory endodomains, including CD28 and 4-1BB (Zhang et al. 2013). The addition of elements such as homing and suicide genes is desirable to improve their safety (Han et al. 2013). The antitumor effects of CAR-modified T cells are well documented. A major breakthrough in the clinical application of CARs was their use in the treatment of hematological tumors. CAR-modified T cell therapy induced complete remission in 45 of 75 adults and children with leukemia (Couzin-Frankel 2013). However, there are fewer clinical trials on the effects of CARs in lung cancer than in leukemia, which could be attributed to the lack of lung cancer-associated antigens, poor T cell trafficking to the tumor site and lower cytotoxicity in the local tumor immunosuppressive microenvironment. Although certain clinical trials are ongoing, early clinical trials assessing anti-CEA T-cell therapy in many cancers, including lung cancer, were terminated because of safety concerns and lack of efficacy or completed without reported results (Table 2).

Limitations of CAR are off-target autoimmune toxicities and possible insertional mutagenesis. For example, CD19-targeted CAR T cells can attack $CD19^+$ normal cells as well as chronic lymphocytic leukemia cells, resulting in a sustained deficiency of $CD19^+$ B cells (Kalos et al. 2011). Attempts to resolve this dilemma should focus on selecting TAAs with high specificity. Molecular targets such as EGFR, VEGFR and CEA are currently used in many kinds of immunotherapy, including vaccines, monoclonal antibodies, bispecific antibodies and gene-modified lymphocytes (Table 2). Most of them are still under research except for certain drugs such as Cetuximab, which has shown beneficial effects in many patients (Pirker et al. 2009). However, these antigens are not only expressed in lung cancer cells, but also other solid tumor cells. Several lung cancer studies have shown that proteins such as KK-LC-1, which is overexpressed on the surface of lung cancer cells, can promote T-cell recognition and killing, indicating that they may become potential targets for immunotherapy of lung cancer (Fukuyama et al. 2006). If further studies can prove their specificity, they may be selected as targets for peptide vaccines or CAR-modified T cells. There are many studies on vaccines and monoclonal antibodies for lung cancer, and some of them are clinical trials (Table 2). Although gene-modified T cells have some targets in common with vaccines, the latter has the advantage of large-scale production, whereas the former has shown slow development. Future work needs to enhance the collaboration between biological companies, medical centers and academic institutions to promote the efficient production of gene-modified T cells.

Discussion

Despite efforts to improve the function of T cells, the results are not as promising as expected. Only a small number of T cells can penetrate tumor tissues and remain functional in the presence of immune escape and lack of high-specificity antigens. Therefore, the selection of high-specificity antigens is important to improve CAR-modified T cells or vaccines. A list of lung cancer-associated antigens may be helpful for the design of single-target or multi-target immunotherapy treatment plans for different patients (Table 2).

The development of drugs capable of restoring or improving T-cell function at the tumor site may be important to obtain better clinical results. For example, decorin, a prototype member of the small leucine-rich proteoglycan family plays a primary role in regulating fibril assembly in vitro and in vivo. It resides in the tumor microenvironment and affects the biology of various types of cancer by directly or indirectly targeting the signaling molecules involved in cell growth, survival, metastasis and angiogenesis. Several recent studies suggested that decorin is a potential cancer therapeutic agent (Zhang et al. 2009; Fiedler and Eble 2009; Goldoni and Iozzo 2008). A natural component of ginger, 6-gingerol, has been widely reported to possess anti-inflammatory and antitumorigenic activities. Administration of 6-gingerol can inhibit the growth of certain types of tumors established by inoculating tumor cells on the flanks of mice, such as B16F1 melanomas, H-1299 human lung cancer, Renca renal cell carcinomas and CT26 colon carcinomas. In tumor-bearing mice, 6-gingerol treatment causes massive infiltration of $CD4^+$ and $CD8^+$ T cells and $B220^+$ B cells, and it reduces the number of $CD4^+Foxp3^+$ regulatory T cells. Furthermore, the $CD8^+$ tumor-infiltrating T cells in 6-gingerol-treated mice strongly express IFN-γ (Lv et al. 2012; Lin et al. 2012; Ju et al. 2012; Lee et al. 2008). So 6-gingerol could be applied in the process of cell reinfusion and could overcome the low targeting of T cells or gene-modified T cells to a certain degree. In addition, the ability of bispecific antibodies to simultaneously recognize

Table 2 Some clinical trials about immunotherapies of lung cancer on different targets

Targets	Categories	Products	Types	ClinicalTrials.gov identifiers
EGFR	Vaccine		NSCLC/SCLC	NCT00003002, NCT00005023, NCT01376505
	Monoclonal antibody	Cetuximab, Panitumumab		NCT00094835, NCT00095199, NCT00097227
	Gene-engineered T cell			NCT00228358, NCT01869166
VEGFR	Vaccine		NSCLC/SCLC	NCT00633724, NCT00673777, NCT00874588
	Monoclonal antibody	Bevacizumab (Avastin)		NCT00193375, NCT00698516, NCT02054052
	Gene-engineered T cell			NCT01218867
CEA	Vaccine	ETBX-011, GI-6207	NSCLC	NCT00004604, NCT00091039, NCT00128622
	Monoclonal antibody			NCT00738452
	Gene-engineered T cell			NCT00004178, NCT01212887, NCT01723306
MUC1	Vaccine	L-BLP25(BLP25), TG4010	NSCLC	NCT00157209, NCT00415818, NCT00960115
MAGE	Vaccine	GSK1572932A, GSK 249553	NSCLC	NCT00020267, NCT00290355, NCT00455572
	Gene-engineered T cell			NCT01694472
GD	Vaccine	1E10 (Racotumomab)	NSCLC/SCLC	NCT00045617, NCT01460472
	Monoclonal antibody	BCG vaccine		NCT00003279, NCT00037713
NY-ESO-1	Vaccine	IDC-G305, CDX-1401	NSCLC/SCLC	NCT00948961, NCT01522820, NCT02015416
Mesothelin	Vaccine	CRS207	NSCLC/SCLC	NCT00585845
	Monoclonal antibody	Amatuximab		NCT00325494, NCT01413451, NCT01521325
	Gene-engineered T cell			NCT01583686
P53 mutant	Vaccine		NSCLC/SCLC	NCT00019084, NCT00019929
Ras mutant	Vaccine	GI-4000	NSCLC	NCT00005630, NCT00019006, NCT00655161
EpCAM	Bispecific antibody	MT110	NSCLC/SCLC	NCT00569296, NCT00635596
gp96	Vaccine		NSCLC	NCT00503568, NCT01799161
WT-1	Vaccine		NSCLC	NCT00398138
PDL-1	Monoclonal antibody	MEDI4736	NSCLC	NCT02000947
IGF-1R	Monoclonal antibody	AMG 479, Cixutumumab, Figitumumab	SCLC	NCT00791154, NCT00869752, NCT00887159
Lewis Y	Monoclonal antibody	Hu3S193	SCLC	NCT00084799
CD56	Monoclonal antibody	BB-10901, IMGN901	SCLC	NCT00346385, NCT01237678
TERT	Vaccine	Vx-001	NSCLC	NCT01935154

a target antigen and an activating receptor such as CD3 on the surface of an immune effector cell makes them potential tools to redirect immune cells to kill cancer cells (Satta et al. 2013; Yamamoto et al. 2012; Cioffi et al. 2012). They provide a simple solution to a complex problem, and some of them are currently in clinical trials (Table 2).

Advances in our understanding of the relationship between lung cancer and the immune system have led to unequivocal successes in the treatment strategies that are currently being widely reported. T cells, the core of cellular immunotherapy for lung cancer, are critical for the genesis, progression and treatment of tumors. In recent years, T-cell-associated immunotherapy for lung cancer has progressed gradually with encouraging results. In clinical practice, the combination of T-cell immunotherapy with cytokines, vaccines and antibodies has effectively improved overall survival in lung cancer patients. Although it is not a curative treatment for lung cancer, T-cell immunotherapy can effectively remove residual tumor cells after resection in early stage lung cancer patients, and it is regarded as a method of treatment if it turns lung cancer into a chronic respiratory disease and maintains patients in stable condition for the advanced stages of lung cancer.

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Conflict of interest We declare that we have no conflict of interest.

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